

DocRE Pipeline Recommendations (NeurIPS-Oriented)

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1. Core Direction

Main framing:

> Iterative Pair-State Relational Refinement under Noisy Supervision

Do NOT frame the work as:

- “GAT + MoE + LoRA + tricks”

Instead:

- Pair-state reasoning is the core novelty

- Everything else supports it

2. Augmentation Strategy

Recommended

Entity-preserving and relation-preserving paraphrase.

Pipeline

original

↓

mask entities

↓

LLM/T5 paraphrase

↓

restore entities

↓

relation consistency filter

Example

Original:

Google acquired YouTube.

Masked:

[HEAD] acquired [TAIL].

Paraphrased:

[TAIL] was acquired by [HEAD].

Restored:

YouTube was acquired by Google.

Important

Filtering is mandatory.

Keep augmented samples only if:

- relation prediction remains consistent
- pair representation remains similar

Recommended filtering

- RE classifier consistency
- embedding cosine similarity
- relation probability stability

Suggested settings

3. Distant Supervision Strategy

Current:

- top-2 cleaned DS relations per entity pair

This is already much better than raw DS.

Recommended schedule

Training philosophy

DS-heavy warmup

→ mixed training

→ supervised refinement

Confidence-weighted DS

Instead of uniform DS weighting:

$$L = L_{\text{sup}} + \lambda(t) L_{\text{ds}}$$

where $\lambda(t)$ decays during training.

Benefits

- learns broad relational priors early
- reduces noisy fitting later
- improves tail relation robustness

4. LoRA Recommendations

Current setup

$q_{\text{proj}} + v_{\text{proj}}$

rank = 64

Potentially overkill for DocRE.

Recommended setup

$q_proj + k_proj + v_proj$

rank = 16–32

Why add k_proj ?

DocRE depends heavily on:

- attention geometry
- entity interaction topology
- evidence routing

k_proj adapts attention compatibility directly.

Suggested LoRA config

Layer selection

Prefer:

- middle layers
- upper-middle layers

instead of:

- adapting every transformer block

5. Graph Reasoning Architecture

Problem with 2-layer GAT

2 layers often under-reason for DocRED.

DocRE frequently requires:

- multi-hop reasoning
- cross-sentence propagation
- mediator entities

Recommended architecture

4-layer relational GAT

+ residual

+ prenorm

+ stochastic depth

Recommended block

LayerNorm

↓

Relational GAT

↓

Dropout

↓

Residual Add

Suggested hyperparameters

6. Pair-State Reasoning (Main Novelty)

Instead of:

entity embeddings $\rightarrow$ classifier

Use:

entity embeddings $\rightarrow$ pair states $\rightarrow$ iterative refinement

Pair-state initialization

P_{ij} initialized from:

- h_i
- h_j
- pair interactions

Example:

$$P_{ij} = \text{MLP}([h_i, h_j, h_i * h_j, h_i - h_j])$$

Triangle reasoning

Main idea:

$$P_{ij} \leftarrow P_{ij} + \sum_k f(P_{ik}, P_{kj})$$

Meaning:

- reasoning through mediator entities
- compositional relation inference
- structural relational propagation

Sparse mediator routing

Do NOT prune entity pairs.

Instead:

- keep entity pair

- sparsify mediator selection

This preserves recall while improving efficiency.

Recycling refinement

Recommended:

- 2–3 refinement iterations

Example:

$P_0 \rightarrow P_1 \rightarrow P_2$

7. Final Recommended Pipeline

LLM Encoder

↓

LoRA(q,k,v)

↓

Entity Embeddings

↓

Pair-State Initialization

↓

4-layer Relational GAT

(residual + prenorm)

↓

Sparse Triangle Reasoning

↓

2-step Recycling Refinement

↓

Relation Distribution

8. Training Schedule

Phase 1

DS-heavy warmup.

Phase 2

Mixed supervised + DS.

Phase 3

Supervised refinement only.

9. Key Positioning for NeurIPS

Avoid presenting:

many disconnected engineering tricks

Instead present:

> A dynamical pair-state relational refinement framework for document-level relation extraction under noisy supervision.